Indic Speech-to-Speech Evaluation Metric Task

Shared Task Overview

Dataset link: TBD **Submission portal:** TBD **Important dates:** TBD

Description

The **Indic Speech-to-Speech Evaluation Metric Task** is a shared challenge focused on developing automatic evaluation metrics for direct speech-to-speech translation systems. Unlike generation-oriented tasks, this challenge asks participants to build models that predict human quality judgments for translated speech. The task is designed for Indic languages and aims to support fair, reproducible, and human-aligned evaluation of speech translation systems.

The benchmark provides train, development, and test splits with human annotations for multiple quality dimensions, including intelligibility, adequacy, and naturalness. Participants must develop systems that predict these scores and the final composite score as accurately as possible. The task is intended to encourage research on robust metric learning for multilingual and morphologically rich speech translation scenarios.

Data

Dataset link: TBD

The task provides benchmark data for selected Indic language pairs, including Telugu-English, Hindi-English, and Telugu-Hindi directions. Each example contains source speech, system-generated translated speech, and human quality scores.

The dataset is organized into standard splits:

- **Train:** Released with human annotations for model training.
- **Dev:** Released with labels for validation and tuning.
- **Test:** Held out for official evaluation; labels are not released.

Each instance includes human scores for:

- Intelligibility
- Adequacy
- Naturalness

Depending on the final benchmark specification, the data may also include metadata such as language pair, utterance ID, or system ID. The main goal is to train an automatic metric that predicts human evaluation scores from the available input.

Baselines

To support participation, the shared task may provide one or more baseline systems. These can include:

- A simple regression baseline using acoustic and textual features.
- An ASR-based quality estimation baseline.
- A multimodal baseline using source speech and translated speech.
- A score-prediction baseline trained directly on the provided human annotations.

Baselines are intended to give participants a reference point for model development and comparison.

Submission

Participants are required to submit predicted scores for the official test set. The submission should contain the model's predicted values for the evaluation dimensions and, if required, the final overall score.

Each team may submit:

- One primary submission.
- Optional contrastive submissions for additional comparison.

Participants should ensure that all predictions are generated in the official test-set order and follow the required submission format.

Test Data

The test split is released separately and used for official evaluation. Human scores for the test set are withheld during the challenge period. The test data may include unseen speakers, domains, or speaking styles to better assess the generalization ability of submitted metrics.

The purpose of the test set is to measure how well systems estimate human judgments on previously unseen speech translation outputs.

Submission Format

The submission format should be simple and machine-readable. A common format is:

- One score per line, or
- A CSV/TSV/JSON file with one row per test instance.

If multiple dimensions are predicted, participants may submit:

- Separate files for intelligibility, adequacy, and naturalness, or
- A single file containing all predicted scores.

The official format should always match the ordering of the released test set.

How to Submit

Participants must submit their system outputs through the official challenge submission process. The submission package should include:

- Predicted test-set scores.
- A short description of the system.
- Any required metadata or identifiers.

Teams should follow the deadline and formatting rules specified by the organizers. Late or non-conforming submissions may not be considered for official ranking.

System Paper

Each participating team should submit a short system description paper. The paper should describe:

- The overall model architecture.
- Input features and training setup.
- Loss functions and score-prediction strategy.
- Any external resources or pretrained models used.
- Validation strategy and experimental details.

The system paper provides context for the submitted results and helps compare approaches fairly across teams.

Evaluation

Systems will be evaluated by comparing predicted scores against human annotations on the held-out test set. The primary evaluation goal is to measure how well a submitted model aligns with human judgments of speech translation quality.

The main dimensions are:

- **Intelligibility:** How clearly the translated speech can be understood.
- **Adequacy:** How well the meaning of the source is preserved.
- **Naturalness:** How fluent and natural the generated speech sounds.

A final composite score may be computed using a weighted combination of the three dimensions:

$$\text{Final Score} = 0.30 \times \text{Intelligibility} + 0.40 \times \text{Adequacy} + 0.30 \times \text{Naturalness}$$

Leaderboard ranking may be based on correlation with human scores, score agreement, or another correlation-based metric defined by the organizers.

Important Dates

The key dates for the challenge should be listed here:

- Task announcement
- Data release
- Baseline release
- Test set release
- Submission deadline
- System paper deadline
- Result announcement
- Workshop presentation

Exact dates can be added once the timeline is finalized.

Organizers

This section can list the organizing team, affiliated institutions, and task chairs. It may also include contributors responsible for data preparation, benchmark design, and evaluation setup.

Contact

For questions related to the challenge, participants can contact the organizers through the official email address or task website.